Insert In Style: A Zero-Shot Generative Framework for Harmonious Cross-Domain Object Composition

Samsung Research

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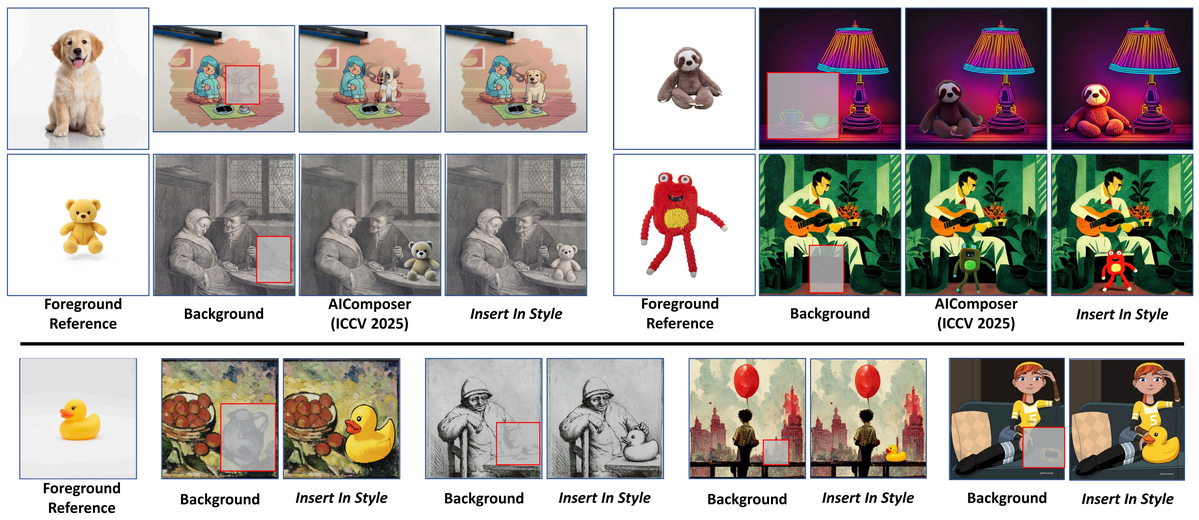
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Overview

Insert In Style: the **first zero-shot generative framework** for cross-domain object composition, **preserving object identity** while **harmonizing it with stylized scenes**.

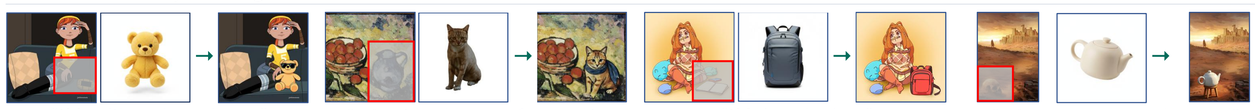
✗ *In-domain composition*: style mismatch ✗ *Training-free blenders*: no native generation ✗ *Online generators*: per-subject finetuning ✓ **Ours: generative · zero-shot · mask-aware**

1 Zero-Shot Cross-Domain Composition

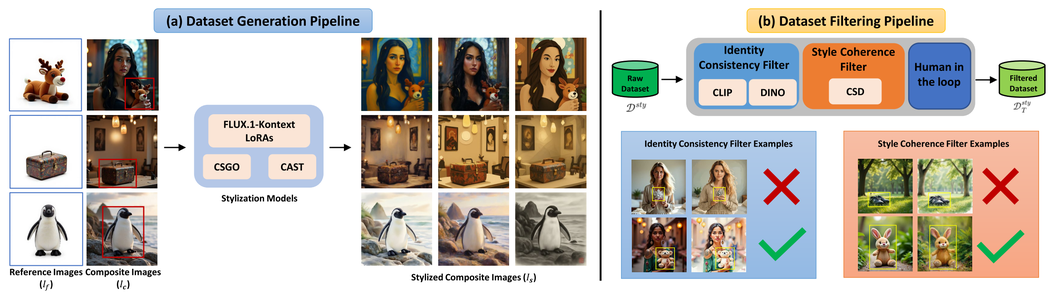


Rows 1–2: AIComposer's *blend-then-refine* corrupts identity with background features; ***Insert In Style*** stays faithful and on-style. **Row 3:** one zero-shot model, four stylized scenes.

Additional Capability: Text-Guided Customization at the Mask



2 Insert In Style Dataset & Bench



- **Generation.** 40k in-domain triplets $\{I_f, I_c, I_m\}$ are stylized with 20 FLUX.1-Kontext LoRAs and with CSGO/CAST on Style30k, giving about 350k composites I_s.
- **Human-in-the-loop filtering.** $\Phi = \{\Phi_{\text{CLIP}}, \Phi_{\text{DINO}}, \Phi_{\text{CSD}}\}$ scores identity and style. Each round removes the lowest decile on any score until expert acceptance saturates:

$$\mathcal{R}_t = \bigcup_{\phi \in \Phi} \{x \in \mathcal{D}_t^{sty} \mid \phi(x) \le Q_{0.1}(\phi)\}$$

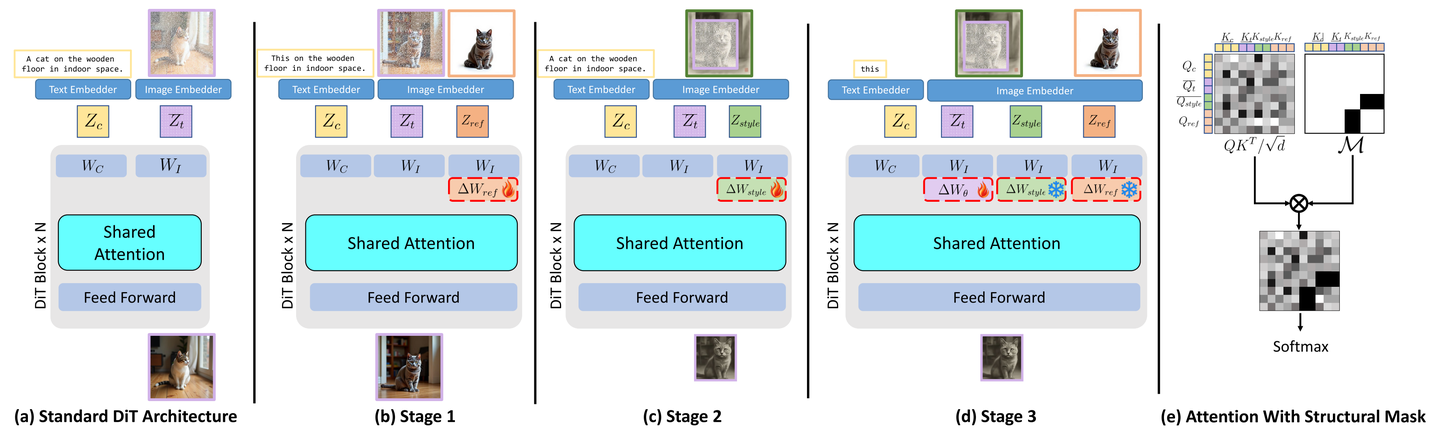
Training Dataset

115k	1,140
curated samples from ~350k	styles LoRA + Style30k

Insert In Style Bench

788	51	25
evaluation pairs largest to date	stylized scenes sketch to neon art	subject concepts pets, food, toys…

3 Proposed Method



Multi-Stage Disentangled Training

On a frozen FLUX.1-dev backbone, **Stage 1** trains the Identity Branch Z_{ref} on Subjects200K and, in parallel, **Stage 2** trains the Style Branch Z_{style} on style-aware inpainting. **Stage 3** freezes both and learns composition with new main-branch LoRA ΔW_θ on the 115k corpus.

Structural Attention Mask

$$S = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}} + \mathcal{M}\right)$$

All branches attend to text Z_c and image Z_t tokens, but $\mathcal{M}$ **blocks** $Z_{ref} \leftrightarrow Z_{style}$. This prevents *concept interference*, so neither condition bleeds into the other.

Prior Preservation

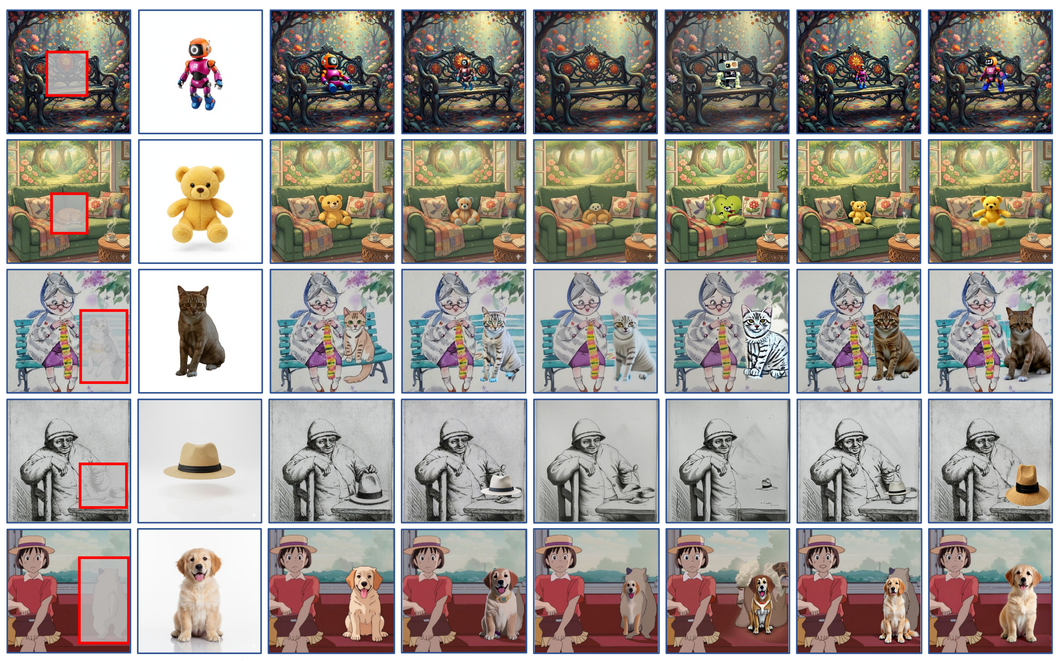
Identity prior: targeted style-context dropout ($p_{drop} = 0.2$) forces object structure to come from the identity reference, not the style condition.

Style prior: a CSD-space loss aligns the inserted object's style with the frozen style model, on the masked region of the one-step denoised estimate.

$$\mathcal{L}_{prior}^{style} = \mathbb{E}\left[1 - \text{CSD}\left(\Psi(v_\theta(Z_t, Z_c, Z_{style}, Z_{ref})),\ \Psi(v_{style}(Z_t, Z_c, Z_{style}))\right)\right]$$

$$\Psi(v) = M \odot \mathcal{D}\left(Z_t + (1-t)\, v\right) \qquad \mathcal{L} = \mathcal{L}_{flow} + \alpha\, \mathcal{L}_{prior}^{style},\ \alpha = 0.2$$

4 Qualitative Comparison



In-domain methods (DreamFuse, AnyDoor) produce style mismatches, and cross-domain methods (AIComposer, TALE, TF-ICON) corrupt subject identity. ***Insert In Style*** is both high-fidelity and harmonious.

User Study

58.6%	74.0%	4.44/5	3.76/5
preferred for fidelity + style next best: 22.7%	preferred for overall aesthetics next best: 13.3%	style coherence next best: 4.08	subject fidelity best: 3.80 (AnyDoor)

5 Quantitative Evaluation

AIComposer Benchmark (367 pairs) · **bold** best, underline second

Method	CLIP-I↑	CSD↑	AES↑	Mean↑
AnyDoor CVPR'24	**0.831**	0.382	0.611	0.608
DreamFuse ICCV'25	0.784	0.458	0.632	0.625
TF-ICON ICCV'23	0.714	0.438	0.584	0.579
TALE MM'24	0.686	**0.495**	0.607	0.596
AIComposer ICCV'25	0.774	0.476	0.644	0.631
Insert In Style	0.779	0.481	**0.655**	**0.638**

Insert In Style Bench (788 pairs)

Method	CLIP-I↑	CSD↑	AES↑	Mean↑
AnyDoor CVPR'24	**0.863**	0.318	0.656	0.612
DreamFuse ICCV'25	0.758	0.449	0.681	0.629
TF-ICON ICCV'23	0.687	0.382	0.661	0.577
TALE MM'24	0.671	0.462	0.695	0.609
AIComposer ICCV'25	0.768	0.430	0.692	0.630
Insert In Style	0.761	**0.466**	**0.697**	**0.641**

Ablation: Training Protocol

Setting	CLIP-I↑	CSD↑	AES↑	Mean↑
Naïve E2E	0.655	0.455	0.668	0.593
w/o Subject pre-train	0.726	0.433	0.678	0.612
w/o Style pre-train	**0.778**	0.399	0.676	0.618
w/o Masked Attention	0.758	0.452	0.690	0.633
Insert In Style	0.761	**0.466**	**0.697**	**0.641**

Ablation: Prior Preservation

Setting	CLIP-I↑	CSD↑
w/o Identity prior	0.731	**0.466**
w/o Style prior	**0.763**	0.432
Insert In Style	0.761	**0.466**

Each prior protects one axis.

Beyond the Main Benchmarks

0.647	0.675	0.641	24.6s
SubjectPlop mean (tied best)	in-domain AnyDoor (best)	vs editors (Qwen-Edit 0.632)	per image (DiT DreamFuse 44.2s)

Conclusion. The multi-stage training protocol, masked-attention architecture and prior preservation losses are designed to manage competing identity and style signals, preventing the *concept interference* common in general-purpose models. Trained on the 115k human-calibrated dataset, *Insert In Style* achieves the best joint identity and style performance and is strongly preferred in user studies.

Project page
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